Discover. Learn. Empower.

Subject: Theory of Computation

Subject Code: 23CST-316/ 23ITT-316

Semester: 6th Batch: 2023

Assignment-3

Last date of Submission: 20 April 2026 Total Marks: 10

*Note: Every student who have score less than 10 marks in MST-2 needs to submit Slow Learner assignment.

SET-A				
Q. No.	**Question**	**Marks**	**CO's**	**BT Level**
1.	Examine the Acceptability of Pushdown automata.	2.5	CO4	4
2.	Implement steps to prove that the Post Correspondence problem is Undecidable.	2.5	CO4	3
3.	Design a TM to accept the language LE={$a^n b^n$ | n >= 1}	5	CO5	6

SET-B				
Q. No.	**Question**	**Marks**	**CO's**	**BT Level**
1.	Differentiate between Deterministic PDA and Non-deterministic PDA.	2.5	CO4	4
2.	Interpret the statement that a recursively enumerable language is said to be recursive.	2.5	CO4	3
3.	Develop language accepted by Turing machine. Construct a Turing Machine for language L = {ww^r | w ∈ {0, 1}}	5	CO5	6

SET-C				
Q. No.	**Question**	**Marks**	**CO's**	**BT Level**
1.	Examine when a problem is said to be decidable and give an example of an undecidable problem.	2.5	CO5	4
2.	Interpret the significance of universal Turing machine.	2.5	CO5	3
3.	Construct PDA which accepts language $L=\{ww^r \mid w\in\{a,b\}^+ \text{ r is any positive integer}\}$	5	CO4	6